A systematic review on the role of the Perfusion Computed Tomography in abdominal cancer

Supplementary Materials

S1. Key terms used in literature search

- "(perfusion computed tomography OR perfusion CT OR CTp) AND abdominal district"
- "(perfusion computed tomography OR perfusion CT OR CTp) AND oncology AND abdomen"
- "(perfusion computed tomography OR perfusion CT OR CTp) AND oncologic disease"
- "(perfusion computed tomography OR perfusion CT OR CTp) AND liver"
- "(perfusion computed tomography OR perfusion CT OR CTp) AND hepatic disease"
- "(perfusion computed tomography OR perfusion CT OR CTp) AND hepatocellular carcinoma OR HCC"
- "(perfusion computed tomography OR perfusion CT OR CTp) AND liver metastases"
- "(perfusion computed tomography OR perfusion CT OR CTp) AND pancreas"
- "(perfusion computed tomography OR perfusion CT OR CTp) AND pancreatic pathology"
- "(perfusion computed tomography OR perfusion CT OR CTp) AND adenocarcinoma"
- "(perfusion computed tomography OR perfusion CT OR CTp) AND kidney"
- "(perfusion computed tomography OR perfusion CT OR CTp) AND renal carcinoma"
- "(perfusion computed tomography OR perfusion CT OR CTp) AND adrenal tumor"
- "(perfusion computed tomography OR perfusion CT OR CTp) AND colon"
- "(perfusion computed tomography OR perfusion CT OR CTp) AND colorectal cancer"
- "(perfusion computed tomography OR perfusion CT OR CTp) AND colon segments AND tumor"
- "(perfusion computed tomography OR perfusion CT OR CTp) AND stomach"
- "(perfusion computed tomography OR perfusion CT OR CTp) AND gastrointestinal system"
- "(perfusion computed tomography OR perfusion CT OR CTp) AND gastric cancer"
- "(perfusion computed tomography OR perfusion CT OR CTp) AND cancer AND survival"
- "(perfusion computed tomography OR perfusion CT OR CTp) AND tumor grade"
- "(perfusion computed tomography OR perfusion CT OR CTp) AND treatment"

- "(perfusion computed tomography OR perfusion CT OR CTp) AND transarterial chemoembolization OR TACE"
- "(perfusion computed tomography OR perfusion CT OR CTp) AND transarterial radioembolization OR TACE"

S2. PRISMA Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	3-4
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	5
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	NA
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	5-6
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	5
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	5-6
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	5-6
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	5-6

Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	5-6
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	6-7
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	NA
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	NA
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	NA
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	NA
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	7
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	9-29
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	30
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	NA
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	NA
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	NA
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	NA
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	29-31

Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	31
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	32
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	32

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S3. Quality Assessment

		Risk of Bias				Applicability		
Study	**Year**	**Patient Selection**	**Index Test**	**Reference Standard**	**Flow and Timing**	**Patient Selection**	**Index Test**	**Reference Standard**
Delrue et al.	2011	Low	Unclear	Low	Unclear	Low	Low	Low
Delrue et al.	2011	Low	Unclear	Low	Unclear	Low	Low	Low
Lu et al.	2011	Low	Low	Unclear	Low	Low	Low	Low
Kanda et al.	2012	Unclear	Unclear	Unclear	Unclear	Low	Low	Low
Khan et al.	2012	Low	Low	Low	Low	Low	Low	Low
Kim et al.	2012	Unclear	Unclear	Unclear	Low	Low	Low	Low
Bai et al.	2014	Unclear	Low	Low	Low	Low	Low	Low
Bayraktutan et al.	2014	Unclear	Unclear	Low	Unclear	Low	Low	Low
Chen et al.	2014	Low	Unclear	Unclear	Unclear	Low	Low	Low
Ippolito et al.	2014	Low	Low	Low	Low	Low	Low	Low
Singh et al.	2014	Unclear	Low	Low	Low	Low	Low	Low
Sun et al.	2015	Low	Low	Low	Low	Low	Low	Low
Sun et al.	2015	Low	Low	Low	Low	Low	Low	Low
Yadav et al.	2016	Unclear	Low	Low	Low	Low	Low	Low
Zongqiong et al.	2016	Low	Low	Low	Low	Low	Low	Low
Mohammed et al.	2017	Unclear	Low	Low	Unclear	Low	Low	Low
Shalaby et al.	2017	Low	Low	Low	Unclear	Low	Low	Low
Aslan et al.	2018	Unclear	Low	Low	Low	Low	Low	Low
Deniffel et al.	2018	Unclear	Unclear	Low	Low	Low	Low	Low
Zaborienè et al.	2021	Low	Low	Unclear	Low	Low	Low	Low

Table S1. Quality assessment using QUADAS-2 tool.

Study	**Year**	**Study participation**	**Study Attrition**	**Prognostic Factor Measurement**	**Outcome Measurement**	**Study Confounding**	**Statistical analysis and reporting**

Petralia et al.	2011	Low	Moderate	Low	Low	Low	Low
Schlemmer et al.	2011	Low	Low	Low	Low	Low	Low
Yao et al.	2011	Low	Low	Low	Low	Low	Low
Curvo-Semedo et al.	2012	Low	Moderate	Low	Moderate	Moderate	Low
Ippolito et al.	2012	Low	Low	Low	Low	Low	Low
Jiang et al.	2012	Low	Moderate	Low	Moderate	Low	Low
Kanda et al.	2012	Moderate	Moderate	Low	Moderate	Moderate	Moderate
Yang et al.	2012	Low	Moderate	Low	Low	Low	Low
Chen et al.	2013	Low	Low	Low	Moderate	Moderate	Low
Morsbahc et al.	2013	Low	Moderate	Low	Low	Moderate	Moderate
Hansen et al.	2014	Low	Moderate	Low	Moderate	Low	Low
Ippolito et al.	2014	Moderate	Low	Low	Low	Low	Low
Nishikawa et al.	2014	Moderate	Moderate	Low	Moderate	Moderate	Moderate
Reiner et al.	2014	Low	Moderate	Low	Moderate	Moderate	Low
Wang et al.	2014	Low	Low	Low	Low	Low	Low
Du et al.	2015	Low	Moderate	Low	Moderate	Moderate	Low
Kaufmann et al.	2015	Low	Moderate	Low	Moderate	Moderate	Moderate
Kaufmann et al.	2015	Low	Low	Low	Low	Low	Low
Lv et al.	2015	Low	Low	Low	Low	Low	Low
Reiner et al.	2015	Low	Moderate	Low	Low	Low	Moderate
Wu et al.	2015	Low	Moderate	Low	Moderate	Moderate	Low
Xu et al.	2015	Low	Moderate	Low	Moderate	Moderate	Low
Marquez et al.	2016	Low	Moderate	Low	Moderate	Moderate	Low
Su et al.	2016	Low	Low	Low	Moderate	Moderate	Low
D'Onofrio et al.	2017	Low	Moderate	Low	Low	Low	Low
Kaufmann et al.	2017	Low	Moderate	Low	Moderate	Moderate	Low
Marquez et al.	2017	Low	Low	Low	Low	Low	Low

Popovic et al.	2017	Low	Moderate	Low	Moderate	Moderate	Low
Shalaby et al.	2017	Low	Moderate	Low	Low	Low	Low
Tamand et al.	2017	Low	Low	Low	Low	Low	Low
Detsky et al.	2018	Moderate	Moderate	Moderate	Moderate	Moderate	Moderate
Ippolito et al.	2018	Low	Moderate	Low	Low	Low	Low
Liang et al.	2018	Low	Low	Low	Low	Low	Low
Nakamura et al.	2018	Low	Moderate	Low	Moderate	Moderate	Low
Ng et al.	2018	Low	Moderate	Moderate	Moderate	Moderate	Low
Shen et al.	2018	Low	Moderate	Low	Moderate	Moderate	Low
Andersen et al.	2019	Low	Moderate	Low	Low	Low	Low
Hamdy et al.	2019	Moderate	Moderate	Low	Moderate	Moderate	Moderate
Lee et al.	2019	Low	Moderate	Low	Moderate	Moderate	Low
Tian et al.	2020	Low	Moderate	Low	Moderate	Moderate	Low

Table S2. Quality assessment using QUIPS tool.